

Fall 23, 4100 Problem Set ~~7~~ 8, Hayden Fuller

Please complete all problems. For any proof questions, please structure your proof similar to those from lecture.

1. Find the spectral decomposition of the self-adjoint matrix

$$A = \begin{bmatrix} 4 & 1+i \\ 1-i & 3 \end{bmatrix}$$

$$\det(A - \lambda I) = 0$$

$$\det \begin{bmatrix} 4-\lambda & 1+i \\ 1-i & 3-\lambda \end{bmatrix} = (4-\lambda)(3-\lambda) - (1+i)(1-i) = (12 - 4\lambda - 3\lambda + \lambda^2) - (2) = \lambda^2 - 7\lambda + 10 = (\lambda - 5)(\lambda - 2)$$

$$\lambda = 5, 2$$

$$u_1 = \frac{\sqrt{3}}{3} \begin{bmatrix} 1+i \\ 1 \end{bmatrix}$$

$$u_2 = \frac{\sqrt{6}}{6} \begin{bmatrix} -1-i \\ 2 \end{bmatrix}$$

$$P_1 = u_1 u_1^* = \begin{bmatrix} 2/3 & (1+i)/3 \\ (1-i)/3 & 1/3 \end{bmatrix}$$

$$P_2 = u_2 u_2^* = \begin{bmatrix} 1/3 & -(1+i)/3 \\ -(1-i)/3 & 2/3 \end{bmatrix}$$

$$A = 5P_1 + 2P_2 = 5u_1 u_1^* + 2u_2 u_2^*$$

2. Let A be hermitian and let λ be a (real) eigenvalue of A . Prove that $\text{null}((A - \lambda I)^2) = \text{null}(A - \lambda I)$.

let $B := A - \lambda I$, then B is Hermitian so $B^* = B$

$\ker(B) \subseteq \ker(B^2)$ since if $Bx = 0$ then $B^2x = B(Bx) = B(0) = 0$

for the reverse, $\langle B^2x, x \rangle = \langle Bx, B^*x \rangle = \langle Bx, Bx \rangle = \|Bx\|^2$

since $B^2x = 0$, the left side is zero, $\|Bx\|^2 = 0$, $Bx = 0$, so $x \in \ker(B)$

combining gives $\ker(B^2) = \ker(B)$, so $\text{null}((A - \lambda I)^2) = \text{null}(A - \lambda I)$

3. Let A be a symmetric, positive definite, $n \times n$ matrix and let B be an $n \times m$ matrix. Prove that $B^T AB$ is positive definite if and only if $\text{rank}(B) = m$.

let $y := Bx \in \mathbb{R}^n$

$$x^T(B^T AB)x = (Bx)^T A(Bx) = y^T Ay$$

assume $B^T AB$ is positive definite.

if $Bx = 0$ then the left hand side is $x^T(B^T AB)x = 0$.

since $B^T AB$ is positive definite, this can only happen when $x = 0$

thus $\ker B = \{0\}$, so B is injective and $\text{rank} B = m$

conversely, assume $\text{rank} B = m$, then $\ker B = \{0\}$

for any nonzero $x \in \mathbb{R}^m$ we have $y = Bx \neq 0$

because A is positive definite, $y^T Ay > 0$

hence $x^T(B^T AB)x = y^T Ay > 0$ for every nonzero x , so $B^T AB$ is positive definite

combining, QED

4. Use linear algebra to find the minimum value of the paraboloid, $f(x_1, x_2) = 2x_1^2 + 4x_1x_2 + 3x_2^2 - 4x_1 + 5x_2 + 8$. Be sure to justify why the value you obtain is indeed a minimum.

$$x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$\text{let } Q = \begin{bmatrix} 2 & 2 \\ 2 & 3 \end{bmatrix},$$

$$b = \begin{bmatrix} -4 \\ 5 \end{bmatrix},$$

$$c = 8$$

$$\text{so } f(x) = x^T Q x + b^T x + c$$

$$\nabla f(x) = (Q + Q^T)x + b = 2Qx + b$$

$$\text{setting } \nabla f = 0 \text{ we get } 2Qx + b = 0, Qx = -b/2$$

$$x^* = \begin{bmatrix} 11/2 \\ -9/2 \end{bmatrix}$$

test $H = \nabla^2 f(x) = 2Q$ check Q is positive definite, $\det Q = 2 * 3 - 2 * 2 = 2 > 0$, so f is strictly convex and x^* is the unique global minimizer

min value evaluate $f(x^*) = -57/4 = -14.25$

5. Consider the Rayleigh quotient, $R(x_1, x_2) = \frac{5x_1^2 + 6x_1x_2 + 13x_2^2}{x_1^2 + x_2^2}$ where $\mathbf{x} \neq \mathbf{0}$. Find a symmetric matrix A such that $R = \frac{\mathbf{x}^T A \mathbf{x}}{\mathbf{x}^T \mathbf{x}}$ and then find the minimum and maximum values of R . Be sure justify why these values are indeed the extreme values of R .

$$A = \begin{bmatrix} 5 & 3 \\ 3 & 13 \end{bmatrix}$$

$$\mathbf{x}^T A \mathbf{x} = 5x_1^2 + 6x_1x_2 + 13x_2^2 \text{ so indeed } R(x) = \frac{\mathbf{x}^T A \mathbf{x}}{\mathbf{x}^T \mathbf{x}}$$

$$\det(A - \lambda I) = (5 - \lambda)(13 - \lambda) - 3^2 = \lambda^2 - 18\lambda + 56$$

$$\lambda = 14, 4$$

because A is symmetric it's orthogonally diagonalizable, $A = Q \text{diag}(14, 4) Q^T$

$$\mathbf{x} = Q\mathbf{y} \text{ gives } R(x) = \frac{\mathbf{y}^T \text{diag}(14, 4) \mathbf{y}}{\mathbf{y}^T \mathbf{y}} = \frac{14y_1^2 + 4y_2^2}{y_1^2 + y_2^2} = 14 \frac{y_1^2}{y_1^2 + y_2^2} + 4 \frac{y_2^2}{y_1^2 + y_2^2}$$

the two fractions are non negative and sum to 1, so $R(x)$ is a convex combination of 14 and 4,

so for all $\mathbf{x} \neq \mathbf{0}$, $4 \leq R(x) \leq 14$ with $R(x) = 14$ when $y_2 = 0$ and $R(x) = 4$ when $y_1 = 0$ therefore the minimum is 4 and maximum is 14.

these are extreme values because for a symmetric matrix the Rayleigh quotient ranges exactly between the smallest and largest eigenvalues.

6. Let A and B be self-adjoint, positive definite matrices. Prove that the eigenvalues of AB are also real and positive.

Let A, B be self-adjoint and positive definite.

Since A is positive definite it has unique Hermitian positive definite square root invertible $A^{1/2}$

$$C := A^{1/2}BA^{1/2}$$

because $A^{1/2}$ and B are Hermitian, C is Hermitian

$$C^* = (A^{1/2}BA^{1/2})^* = A^{1/2}BA^{1/2} = C$$

C is positive definite for any nonzero x $x^*Cx = (A^{1/2}x)^*B(A^{1/2}x) > 0$,

since $A^{1/2}x \neq 0$ and B is positive definite

all eigenvalues of C are real and positive

AB is similar to C , with $S := A^{1/2}$,

$$S^{-1}(AB)S = A^{1/2}ABA^{1/2} = A^{1/2}BA^{1/2} = C$$

$$\sigma(AB) = \sigma(C)$$

therefore every eigenvalue of AB is an eigenvalue of C so it real and positive

thus the eigenvalues of AB are real and positive

QED